01 / Digital implementation delivered

US11405248B2 - FPGA based system for decoding PAM-3 signals

A complete portable, bounded-memory digital reconstruction from two deserialized 8-bit ADC streams to channel-tagged PAM3 host records. Created 4 October 2026. The supplied 19-page patent is included and the reconstruction choices are explicit.

Measured verification	Result
RTL configurations / actual executions	8 / 16 - all passed
Core + paired analyzer sample checks	830,640
Exact output symbol comparisons	73,206
Channel-capture test instances	1,440
XC7 mapped-netlist simulations	Both passed; 23,574 channel samples
Hardware latches after optimization	0 in checked configurations

Implemented boundary

RTL includes literal optional DC correction, slope slicing, spike removal, both edge filters, sampling-point reconstruction/filtering, symbol extraction, dual-channel synchronization, sparse symbol packing, a registered host arbiter, and a separate overflow-reporting capture FIFO. Python parses the binary host record stream.

Physical work still required

No board-specific ADC PHY/clock-domain crossing, coupler/DVGA circuit, USB/DMA controller, Ethernet PCS decoder or board bitstream is included. Vivado was not available. Yosys mapping and functional simulation do not establish 800 MS/s timing, power or real-channel accuracy.

Primary source: supplied source_patent.pdf, US 11,405,248 B2, granted 2 August 2022; figures 1-4 and printed columns 5-8. File/figure interpretations: docs/PATENT_ASSUMPTIONS.md.

02 / Streaming architecture

Stage	Bounded state / role
ADC / slope	Previous ADC and decoded level; optional 127-byte causal history
Spike filter	Five original level positions plus boundary-valid bits
Edges / anchors	Previous edge and direction; four pending edges; 32-bit anchor ring
Sampling / symbol	32-entry level ring; four candidate and essential bits; 64-bit indices
Per-channel packer	32+LANES two-bit symbols plus one output record register
Host output	Round-robin arbitration, one 128-bit registered record

Why the lookahead is 20 sample positions

Filtered position t-2 uses two future raw levels. A must midpoint is at most seven positions before its right edge; the second edge filter needs four more positions. The anchor is known within 13 input positions after that midpoint. Candidate evaluation at t-16 leaves margin. The final four-position candidate filter emits the selected point at t-20.

A 32-entry filtered-level ring is sufficient: a final symbol read occurs 18 positions after the filtered value was written, before reuse. Total memory does not grow with capture length. The optional DC window remains fixed at 128 samples.

Lanes and stalls

LANES=1 or 8 advances consecutive real samples in chronological order. Eight lanes unroll dependent logic in a single clock; this is not an eight-stage clock pipeline. A blocked output freezes sample-time. Paired input transfers occur only when both channels are ready, while decisions and outputs remain independent.

Capture termination

A final ADC beat schedules 20 invalid virtual positions to flush boundary decisions. These do not update ADC/DC/adaptive state. The terminal output must be accepted before history is reset and the capture ID increments. Empty captures still have a terminal record. With no terminal beat, processing runs continuously.

03 / Patent semantics and assumptions

Element	Delivered interpretation
Four thresholds	Plus=140, Minus=95, ZeroPlus=160, ZeroMinus=80
Rising/falling	Strict comparisons; equality or constant ADC holds the previous level
Spike	Fill isolated original runs of width 1 or 2 only when neighbours agree
Must transition	Opposite consecutive edge directions, separated by <=12 samples
Must midpoint	floor((left_edge+right_edge-1)/2), inside the intervening plateau
First edge filter	Must edge or >=9 samples from preceding raw edge; first edge retained
Second edge filter	Drop edge if a must edge appears within the next four positions
Sampling	Start phase 6, step 12, restart at retained must midpoint
Candidate filter	Keep essential points; drop a nonessential point if another is <=4 later

Literal DC ambiguity preserved

The printed correction is y=x-MIN+95, with widened arithmetic and strict y<255 gate. Because the causal window includes x, y is at least 95 when correction is active. The interpreted falling decision for -1 requires y<95. These conditions cannot coincide while correction is active. The patent does not resolve the threshold naming/equality ambiguity; the disclosed bypass mode is therefore the default. The literal mode remains implemented and tested.

Adaptive option is an engineering reconstruction

Q8 integer means start at 60/127/194. Completed 12-sample blocks update classes with at least three samples, using a 1/8 EWMA and floor arithmetic. Threshold ratios are 13/67, 32/67, 33/67 and 47/67, followed by half-up rounding and clamping. No transmitter labels or fitted transmitter phase enter the decoder. This is not an undisclosed vendor algorithm; it also differs in rounding from the earlier floating-point simulation.

Detailed source mapping: source_patent.pdf columns 5-6 (ADC/DC/slicing), column 7 and figures 2/4F/4G (spikes and edges), columns 7-8 (sampling), figure 1/column 8 (host boundary).

04 / Actual RTL verification

Lanes	DC	Adapt Q8	Core symbols	Host symbols	Result
1	0	0	3098	6202	PASS
1	1	0	2984	5967	PASS
1	0	1	3090	6175	PASS
1	1	1	3032	6055	PASS
8	0	0	3098	6202	PASS
8	1	0	2984	5967	PASS
8	0	1	3090	6175	PASS
8	1	1	3032	6055	PASS

Each configuration tested 60 captures per channel. Total core checks: 276,880; dual-analyzer channel sample checks: 553,760. These are repeated sample exposures across builds, not that many independent physical measurements.

What was checked

Every corrected ADC code, raw decoded level and spike-filtered level; every selected output index and symbol; exact symbol counts, terminal captures, and stable payloads under backpressure. The integrated analyzer records are parsed and compared per channel/capture. The oracle uses independent offline run/edge/candidate lists rather than RTL pipeline state.

Input coverage

Empty and 1-33-sample tails; constant ADC codes including exact thresholds and extremes; strict-threshold sequences; ramps; long runs; one/two/three-sample pulses; random lengths around 128/256/1024 boundaries; and synthetic PAM3 at three noise settings. Partial vectors also occur inside captures. Random input bubbles, random sink stops and 180-clock output-stop bursts exercise finite buffering.

Interface and structural checks

Capture FIFO: 6,000 clocks, 2,606 ordered read checks and 1,940 explicitly counted drops. Core protocol test: malformed counts, reset and empty/tiny intervals; 100 consecutive eight-sample beats accepted with a ready sink. Eight Python parser tests cover legal records and malformed/truncated captures. All eight RTL parameterizations and the capture FIFO passed lowered-netlist connectivity/no-latch checks.

Evidence: results/verification/validation.json, results/interfaces/validation.json and results/structure/validation.json. Simulation suite: Icarus Verilog 12.0; Python 3.12.14.

05 / Synthesis and mapped-circuit checks

Digital two-channel top	LUT cells	FF cells	CARRY4	Mapped cells
L1_DC0_A0	2,794	1,801	378	5,353
L8_DC0_A0	21,908	3,863	2,676	33,937

Tool: Yosys 0.33 (git sha1 2584903a060). Target family: generic Xilinx 7-series (XC7), with I/O/clock buffer insertion disabled. Counts include both decoder channels, both packers and host mux. The separate capture FIFO is not part of these counts. LUT cells are counted primitives, not a device occupancy or post-route slice estimate.

Flow and actual circuit simulation

The synth_xilinx flow omits optional SAT resource sharing. Explicit signed PAM3 and control-state encodings prevent automatic FSM extraction from enumerating a large unrolled input table. Check assertions passed with no optimized hardware latches. Mapped Verilog and EDIF are included.

Mapped configuration	Channel samples	Symbols	Result
L1_DC0_A0	20,404	1,871	PASS
L8_DC0_A0	3,170	297	PASS

Both mapped circuits were executed with the XC7 functional primitive library and independently generated inputs, including output stalls. These are functional netlist tests without timing delays. Structural checks cover DC/adaptive options; only the fixed/DC-bypass 1- and 8-lane tops are technology-mapped in this delivery.

No timing claim

The 100 MHz clock / eight-lane setting corresponds arithmetically to an 800 MS/s input target. It is not measured or guaranteed. Eight-lane dependencies, window extrema and optional constant divisions can create long paths. Vivado synthesis/place/route and WNS/TNS review on the selected part remain required. No board or physical USB throughput was measured.

06 / Host records and capture integrity

Bits	Field
63:0	Up to 32 two-bit symbols, earliest at bits 1:0
70:64	Valid count, 0..32
71	Channel 0=A/master, 1=B/slave
72	Terminal record of this channel capture
104:73	32-bit capture ID, initially zero
127:105	Reserved zero

Codes: 11=-1, 00=0, 01=+1; 10 is illegal. Unused payload bits are zero. Binary files contain 16 little-endian bytes per record. A zero-count record is legal only at capture end. Compact records retain symbol order, not sample indices; the direct core interface retains exact 64-bit indices.

```
python3 host_decode.py results/example_host_records.bin \
  --csv results/example_decoded_symbols.csv \
  --summary results/example_decoded_summary.json
```

Capture buffer and losses

The optional FIFO is single-clock, depth 2^LOG_DEPTH beats. Defaults give 256 paired eight-sample beats (2,048 samples/channel). Full-buffer arrivals that cannot be stored set a sticky overflow and increment a saturating dropped-beat counter. Any such loss invalidates the affected acquisition: reset/restart rather than interpreting it as continuous. A multi-clock ADC requires a separate verified asynchronous FIFO.

Bandwidth arithmetic

At nominal 66.667 MBd, two channels of two-bit symbols carry about 33.333 MB/s. Full 128-bit records for 32 symbols require about 66.667 MB/s before transport overhead. Short captures add overhead. The two-channel 8-bit 800 MS/s raw boundary is 1.6 GB/s. These budgets assume nominal symbol cadence and are not throughput measurements.

07 / Vivado and physical integration

The scripts are ready for use with an installed Vivado and an actual device. The default xc7a200tfbg484-2 is only an example part. No pin locations are invented. The OOC example constrains a 10 ns clock and abstract 2 ns input/output delays; these are not ADC LVDS timing constraints.

```
vivado -mode batch -source vivado/create_project.tcl \
  -tclargs xc7a200tfbg484-2 8 0 0

vivado -mode batch -source vivado/synth_ooc.tcl \
  -tclargs xc7a200tfbg484-2 8 0 0 1
```

Next physical milestone	Acceptance evidence
Choose board / ADC / analog parts	Datasheets, channel range, parallel output width and clock plan
ADC PHY and CDC	Vendor capture/deserialization, pin/I/O constraints, timing and continuity checks
Implementation timing	Placed/routed WNS/TNS, constrained-path coverage and resource review
Transport adapter	Byte-correct 128-bit records, sustained bandwidth and overflow handling
Analog calibration	Known PAM3 patterns and measured gain/DC/noise/channel separation
Protocol layer if required	PCS/descrambler and Ethernet decoding against valid known frames

Capture intervals are not Ethernet packets. The patent gives a host protocol-analysis boundary, not enough detail to reconstruct a vendor USB driver or complete Ethernet PCS. These physical and protocol modules require explicit component/standard selections and real-device testing.

08 / Reproduction and deliverable map

```
python3 -m pip install -r requirements.txt
python3 verify_hdl.py
python3 verify_interfaces.py
python3 check_structure.py --yosys yosys
python3 run_synthesis.py --lanes 8
python3 verify_netlist.py --lanes 8 \
  --cells /usr/share/yosys/xilinx/cells_sim.v
```

Path	Purpose
rtl/	Six portable synthesizable SystemVerilog modules
reference/	Numerical fixed model and integer adaptive oracle
tests/	Core, analyzer, FIFO and protocol testbenches; parser tests
host_decode.py	Binary record parser and CSV export
vivado/	Project/OOC scripts and example constraints
results/	Actual validation logs, mapping statistics, mapped circuits and example capture
docs/	Architecture, interfaces, reconstruction assumptions and supplied patent
README_FA.md	Persian startup/integration guide
SHA256SUMS.txt	Included-file integrity checks

Interpretation of a passing result

The delivered digital reconstruction has repeatable exact equivalence tests and actual synthesis artifacts. Passing tests establish agreement for the stated inputs and implemented semantics. They do not prove undisclosed vendor semantics, all possible inputs, a physical sample clock, real-channel SER or an automotive analyzer certification.

The existing functional simulation package remains a separate earlier milestone. This delivery adds the complete streaming digital core and host path. Review the documented literal-DC ambiguity and the integer adaptive update before changing thresholds or integrating a real ADC.